

Worksheet 2 – K-means method

Suppose we measure two variables X_1 and X_2 for each of four items A, B, C and D. The data are given in the following table

Item	Observations	
	x_1	x_2
A	5	3
B	-1	1
C	1	-2
D	-3	-2

(a) Use MacQueen's K-means method to divide these items into 2 clusters such that the items within a cluster are closer to one another than they are to the items in different clusters. Start by randomly selecting 2 rows (any 2 items out of the 4) as the initial centres. For this exercise, let's pick A and C as initial centres so we will all obtain the same answers.

Note: As programmers, students should develop the habit of looking at R documentation rather than rely solely on notes and the internet to code. Take a look at R documentation and read the *centers* argument carefully to understand how to implement proper code for this problem.

(see next page for part b)

(b) Calculate the total within cluster sum of squares for the result in part (a)